IB 514: Scientific Writing

All course slides strung together

March 20, 2026

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Overview

figs/sciwriting.png

Core Principles

Being a scientist is being a writer

XYZ

Website & Schedule

https://github.com/ScientificWriting/

Accountabilibuddies

- Weekly check-in
  - *goals/deadlines, accomplishments, sticking-points*
  - *final 5 min. of every 2nd class*
- Randomized pairing
  - *re-randomized week 6*

What you'll need

Software

- X

Accounts

- X

figs/phd101212s.png

Project Proposals

Primary Goal

To develop more effective writing skills.

Secondary Goal

To make significant progress on a manuscript.

Let's get started!
.../classes/ProjectProposal

We're all learning...

- Be kind to yourself
- Support each other

Integrity

Doing the right thing, even when no one is watching.

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Session Objectives

- Build sustainable daily writing routines
- Design protected writing time
- Structure deadlines strategically
- Improve drafting and workflow practices
- Diagnose and intervene on writer's block
- Develop meta-cognitive awareness

Daily Writing as a System

Core Principle: Writing is a repeatable cognitive practice.

- Frequency $>$ intensity
- Define task before session begins
- Protect high-cognition hours

Some Models of Daily Writing

1. **Consistency Model**
   - 30–60 min daily
   - Same time, same location
2. **Sprint Model**
   - 25-min focused intervals
   - 2–3 per day
3. **Cognitive Load Model**
   - Morning: argument development, drafting new sections
   - Midday: data checks, figure refinement
   - Late day: formatting, references, minor edits
4. **Paragraph Commitment Model**
   - 1 paragraph/day
   - Stop after completion

Concrete Output and Task Decomposition

- 1 paragraph/day $\rightarrow$ 5 per week
- 300 words/day $\rightarrow$ 6,000 per month
- Draft Results before Introduction

Task Decomposition

- Instead of "Revise Discussion"

  $\rightarrow$ Cut 300 words

  $\rightarrow$ Clarify paragraph 3 claim

  $\rightarrow$ Add 2 citations on stability theory

Protected Writing Time

Protected time is:

- Scheduled
- Visible
- Non-negotiable

Treat it as a meeting with yourself.

Implementation Strategies

- Block 60–90 min during peak cognitive hours
- Close email + messaging platforms
- Define task in calendar description
- Begin immediately (no warm-up browsing)
- Start with 2–3 blocks/week; increase after consistency

Major vs. Minor Deadlines

Nested Planning Structure

Major deadlines determine direction.
Minor deadlines determine daily behavior.

Major Deadlines

- Journal submission
- Thesis chapter defense
- Circulate full draft to co-authors

Minor Deadlines

- Draft study system section of Methods by April 1
- Figure 2 drafted by Tuesday
- Circulate Introduction in 10 days
- Draft Abstract by Friday

Examples of Minor and Micro Deadlines

Examples within a 4-week revision window:

- Day 3: Rewrite Results subsection 1
- Day 5: Reduce Discussion by 400 words
- Week 2: Complete robustness checks
- Week 3: Internal logic review of Introduction
- Week 4: Format references

Micro-Deadlines

- Clarify claim of paragraph 4 (30 min)
- Verify statistical assumption 2
- Insert 3 missing citations
- Rewrite topic sentences in Results

Minor goals should be specific, time-bound, and measurable.

Workflow as a Pipeline

Ideas $\rightarrow$ Notes $\rightarrow$ Analysis $\rightarrow$ Figures $\rightarrow$ Manuscript

Goal: reduce switching costs.

Explicit Transition Points

- Notes $\rightarrow$ Analysis: Research question formalized
- Analysis $\rightarrow$ Figures: Model validated + assumptions checked
- Figures $\rightarrow$ Manuscript: Figure narrative drafted in bullet form
- Manuscript $\rightarrow$ Circulation: Logical coherence pass completed

Definition of Done Examples

- Analysis complete = validated + documented
- Figure complete = caption explains inference
- Section draft complete = each paragraph has explicit claim

Cognitive Energy Management

High-energy tasks:

- Argument construction
- Structural revision

Medium-energy tasks:

- Sentence rewriting

Low-energy tasks:

- Formatting
- Reference checks

Align task to cognitive state.

Draft Before You Feel Ready

Write imperfect prose intentionally

"Write drunk, edit sober." – Hemingway

Separate drafting from editing.

- Draft full section quickly
- Revise next day

Reverse Outline in Bullet Points

Reverse outlining

Extracting the logical structure after drafting.

Process

- Reduce each paragraph to one bullet:

  "What claim does this paragraph make?"
- Examine logical order.

Example

Paragraph claim:

Species diversity increases stability under environmental fluctuation.

Reverse outline bullet:

Diversity buffers variance via asynchronous population responses.

If bullets do not form a coherent argument, restructure.

Diagnosing Writer's Block: Underlying Causes

Writer's block is usually a symptom, not a primary problem.

1. **Cognitive Causes**
   - Argument is not fully specified
   - Paragraph lacks a clear claim
   - Logical steps between ideas are missing
   - Methods or assumptions not fully understood
2. **Structural Causes**
   - No outline or unstable outline
   - Section scope too broad
   - Unclear transition between sections

Diagnosing Writer's Block: Underlying Causes

Writer's block is usually a symptom, not a primary problem.

3. **Process Causes**
   - Editing while drafting
   - Task too large ("Revise paper")
   - No defined next action
4. **Psychological Causes**
   - Perfectionism
   - Fear of evaluation (advisor, reviewers)
   - Avoidance of cognitively demanding work

Writer's Block

A signal that something specific needs clarification.

Interventions: Matching Strategy to Cause

If the argument is unclear:

- Write the paragraph's claim as a single sentence.
- Ask: What must be true for this claim to hold?
- Sketch logic on paper before drafting prose.

If the task is too large:

- Reduce to a 15-minute goal.
- Replace "Revise Discussion" with "Rewrite paragraph 2 for clarity."

If perfectionism is interfering:

- Draft in bullet points.
- Use timed freewriting (10 minutes, no deletion).
- Write a deliberately "bad" version.

Interventions: Matching Strategy to Cause

If fatigue is the issue:

- Switch to a lower-cognitive task (references, figure labels).
- Stop and resume during peak cognition.

If stuck mid-paragraph:

- Insert bracketed placeholders: [CITE], [EXPLAIN MECHANISM], [CHECK RESULT]

Momentum restores clarity more reliably than rumination.

Meta-Cognitive Writing

Strong writers monitor:

- When they write best
- How long tasks take
- What triggers avoidance
- What improves clarity

Practical Monitoring Systems

Writing Log

- Date
- Duration
- Words added/removed
- Task completed

Example Log Entry

- 8:00–8:45 AM
- Drafted 2 Results paragraphs
- Stalled at interpretation step
- Insight: need clearer model summary before Discussion

Closing Reflection

Write down:

- One protected writing block
- One minor deadline for this week
- One workflow transition to clarify
- One drafting experiment

Writing productivity is designed, not discovered.

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Framing the Session: Why Workflows Matter

Central Premise: High-quality writing emerges from high-quality systems, not last-minute effort.

Three Guiding Questions

1. How can I reduce friction between ideas and written text?
2. How can I prevent structural chaos as projects grow?
3. How can I ensure my workflow supports collaboration and revision?

Session Objectives

- Design a scalable project structure
- Identify points where inefficiency may emerge
- Evaluate writing tools strategically

Key Insight: Workflow design is a methodological decision

It shapes clarity, reproducibility, and speed.

Overview

Project Structure

Recommended Folder Architecture

- `/notes` – reading notes, idea sketches, meeting notes
- `/data`
  - `/raw` (read-only)
  - `/processed`
- `/code` – scripts, functions, notebooks
- `/results` – exploratory outputs, intermediary results
- `/tables` - publication-ready tables[1]
- `/figures` – exported publication-ready figures
- `/manuscript` – bibliography, drafts and versions[2]

[1] Exported if using LaTeX

[2] If not using version control software

Project Structure

Best Practices

- Explanatory README file in root directory
- Separate raw and cleaned/processed data
- Separate exploratory from finalized results (figs and tables)
- Consistent naming conventions

File Naming Conventions

- Consistent ordering:
  - `survey_raw_2025.csv`
  - `survey_cleaned_2025.csv`

  Consistent capitalization:
  - `Upperstart` vs. `lowercase`
  - `CamelCase` vs. `start_end`
- Use descriptive, machine-readable names:
  - Avoid: `final_real_final2.docx`
  - Prefer: `results/regression_modelA.csv`

Version Control

Why Version Control?

- Prevents loss of work
- Eases identification and fixing of problems/bugs
- Avoid concern over losing removed text
- Enables collaboration
- Documents intellectual development

Approaches

- Manual versioning
  - Use ISO dates: `2026-02-15_analysis.R`
  - Version numbers: `manuscript_v03.tex`
  - Coauthor comments: `manuscript_v03_MN.tex`
- Cloud version history (synced folders, e.g., Dropbox, Box)
- Git-based workflows (e.g., Git with GitHub)

Overview

Writing Workflow: From Ideas to Manuscript

Linear model (non-existent):

$$\text{Ideas} \rightarrow \text{Notes} \rightarrow \text{Data} \rightarrow \text{Code} \rightarrow \text{Figures} \rightarrow \text{Manuscript}$$

Key Insight: The workflow is iterative, not strictly linear.
Most time is spent revising.

Common Transition Points

- Notes $\rightarrow$ structured research questions
- Code $\rightarrow$ validated results
- Clarity $\rightarrow$ candidate figures
- Figures $\rightarrow$ narrative refinement

Reminder:
Draft sections early (especially Methods and Results).

Pre-Drafting Structural Design

Core Principle:

Design structure deliberately; diagnose structure explicitly.

Storyboarding

- Construct the argument visually before drafting prose.
- Create one slide, page, or card per figure or major claim.
- For each, specify:
  - The core claim (1 sentence)
  - The evidence shown
  - Why it matters for the overall argument
- Arrange in logical order to test narrative flow.
- Identify missing transitions or unsupported claims early.

Goal: Ensure overall logical coherence before drafting prose[3]

[3] But remain open to change!

Post-Drafting Structural Diagnosis

Core Principle:

Design structure deliberately; diagnose structure explicitly.

Reverse Outlining

- Extract 1–2 bullet-point claims per paragraph.
- Ignore prose quality; focus only on argumentative function.
- Ask:
  - Does each paragraph have a clear purpose?
  - Does the sequence of bullets form a coherent argument?
  - Are there redundancies or logical gaps?
- Revise structure before revising sentences.

Goal: Ensure both internal and overall logical coherence after drafting prose.

Overview

Writing Tools

Word / Pages / Google Docs

Pros:

- Widely used; track-changes
- Real-time cloud collaboration

Cons:

- No or poor version control
- Formatting instability

LaTeXw/ Git, Overleaf, Prism

Pros:

- Plain text
- Superior math, refs, tables
- Auto-numbering & insertion

Cons:

- Learning curve
- Harder for collaborators
- Cloud-based track-changes/ commenting

Markdown

Pros:

- Lightweight, plain text
- Future-proof format
- Easy code integration

Cons:

- Toolchain setup required
- Limited formating & image handling

Scrivener

Pros:

- Research & reference binder
- Snapshots for safe revisions
- Great for nonlinear drafting

Cons:

- Not for collaborative work
- Proprietary file format
- Export cleanup required

Additional Tools

Dictation Tools

e.g, MS Speech Recognition, macOS Dictation, Otter.ai (300 min./mo.)

Pros:

- Rapidly capture ideas, reduce typing fatigue
- Lowers activation energy to overcome writing inertia

Cons:

- Substantial editing (typos, grammar, clarity, formatting)
- Misinterpretation of technical or domain-specific terms

Large Language Models (LLMs)

e.g., ChatGPT, Claude, Scite, Elicit, Prism, and many more

Pros:

- Assist in drafting sentences, paragraphs, and structured outlines
- Suggest structural improvements, phrasing, and organization
- Literature summaries or conceptual scaffolds

Cons:

- Risk of (over)reliance, inhibiting original thought
- Factual inaccuracies, limited domain knowledge and literature access
- Generic or formulaic writing
- Environmental impact

Overview

Conceptual and Data Figures

Conceptual Figures

- Clarify theoretical frameworks
- Summarize complex relationships

Data Figures

- Present core results
- Should be interpretable independently

Manual vs. Code-Based Creation

- Manual (e.g., Illustrator): high aesthetic control
- Code-based: reproducible and scalable

File Formats Commonly Requested

- PDF – typically sufficient
- TIFF – high-resolution images
- EPS – vector graphics
- Figure resolution (e.g., 300–600 dpi)

Overview

Workflow Integration

Align structure, tools, and formatting strategy.

- Organize for clarity early to reduce friction later.
- Choose tools that match writing needs.
- Separate development and writing from formatting.

Goal:

Efficient, transparent, and reproducible writing workflow.

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Audience Identification

- Know your target audience:
  - General scientific audience
  - Specialists vs. interdisciplinary researchers
  - Theoreticians and/or empiricists
- Audience shapes decisions about:
  - Terminology and background explanation
  - Context and significance framing
  - Level of technical detail
  - Visual presentation choices (e.g., conceptual figs)
- Tailor writing to match audience expertise and interests
- Consider what audience knows vs. needs to know

Journal Selection

- Identify target journals *before* you start writing
- Rank potential journals:
  - Aspirational journals (high impact, broad audience)
  - Realistic journals (good fit for topic and scope)
  - Backup journals (alternative options if rejected)
- Consider:
  - Similarity of published articles to your work
  - Journal scope and audience
  - Open access policies (and costs)
  - Impact factor and prestige (with caution)
- If you're not confident
  - Check target journal requirements for alignment
  - Review recent issues

Signposting

- Guide readers through your argument with effective signposting
- Use transitions and connectives:
  - Topic sentences that signal direction
  - Transitional phrases linking paragraphs and sections
  - Explicit statements of relationships between ideas
- Employ structural markers:
  - Clear section headings and subheadings
  - Summary sentences to recap key points
- Help readers anticipate and understand:
  - What comes next
  - How ideas connect
  - Why information matters

Structural Coherence

- Maintain logical flow throughout the paper
- Structure provides the "roadmap" for readers
- Key aspects:
  - Logical progression of ideas within sections
  - Clear connections between sections
  - Motivation established before presenting methods
  - Results presented before interpretation
  - Conclusions grounded in findings
- Ensure each section connects to and builds on previous

IMRaD (Introduction–Methods–Results–Discussion)

1. **Methods** first (most structurally constrained).
2. **Results** second (figure-driven; claim-evidence alignment).
3. **Introduction** after results clarify true contribution.
4. **Discussion** last to interpret, generalize, and situate.

Introduction (Why?)

- Move from broad problem to specific knowledge gap.
- Critically evaluate literature (don't summarize passively).
- End with explicit research questions, hypotheses, or key finding.

Methods (How?)

- Provide enough detail for replication.
- Use clear subheadings aligned with research questions.
- Justify key analytical decisions.

Results (What?)

- Organize around figures/tables.
- Lead each subsection with the main finding, align with Methods sections
- Avoid interpretation beyond minimal context.

Discussion (So what?)

- Interpret findings relative to hypotheses.
- Compare with prior literature.
- Address limitations and articulate broader implications.

Methods-at-the-End Papers

Design a Figure-Driven Narrative

- Determine the logical sequence of core results.
- Ensure each figure advances the central claim.
- Remove peripheral analyses from main text.

Front-Load Essential Context

- Provide minimal methodological orientation in Results.
- Define key variables and datasets before presenting outcomes.

Write Results as a Coherent Argument

- Each subsection should answer a focused question.
- Maintain conceptual momentum between figures.

Develop Detailed Methods Section at End

- Provide full reproducibility detail.
- Use clear subsections for sampling, measurements, analyses.
- Anticipate reviewer questions and address proactively.

Key: Requires exceptionally clear storytelling

Literature Reviews

- **Define scope precisely**
  - What question is the review answering?
  - What time period, disciplines, or methodologies are included?
- **Develop a transparent search strategy**
  - Databases used, search terms, inclusion/exclusion criteria.
  - Consider a flow diagram for study selection.
- **Organize by conceptual structure (not chronology)**
  - Thematic clusters
  - Competing hypotheses
  - Methodological approaches
- **Synthesize within sections**
  - Compare studies directly.
  - Highlight agreements, disagreements, and methodological drivers.
- **Conclude with implications**
  - Identify unresolved questions.
  - Clarify future research directions.

Synthesis

- **Begin with a conceptual problem**
  - What tension, paradox, or fragmentation exists?
- **Abstract upward**
  - Move from specific findings to higher-level principles.
- **Integrate across domains**
  - Combine theories, datasets, or methods not typically linked.
- **Develop a unifying framework**
  - Conceptual diagram or model.
  - Clear definitions of key constructs.
- **Demonstrate explanatory gain**
  - Show how the synthesis resolves contradictions or generates predictions.

Data Papers

- Clarify dataset purpose and potential reuse scenarios.
- Provide full documentation:
  - Sampling design, instrumentation, preprocessing steps.
  - File structure and variable-level metadata.
- Demonstrate quality control
  - Validation checks, error screening, uncertainty description.
- Include example use cases
  - Illustrative analyses showing dataset functionality.

Methods, Results & Supplementary Materials

Methods and Results

- Sufficient detail for general understanding & reproducibility
- Balance completeness with readability
- Use parallel subheadings to organize

Supplementary Materials

- Use strategically to keep main text focused and concise
- Main text must remain complete and independent
- Include:
  - Extended methods, secondary or alternative analyses
  - Statistical analysis details or results tables
  - Details and results that are not essential to main narrative

Integration of Equations

- Introduce equations before presenting them
- Interpret equations after presenting them
- Equations (symbols) are words in a sentence, not standalone objects (Use appropriate grammar and punctuation)
- Guidelines:
  - Define all variables and parameters
  - Explain the logic and significance of equations to broad audience
  - Follow standard (or explain) mathematical notation
- Keep equations in the main text only if essential to narrative[4]

[4] See Fawcett & Higginson, *PNAS* 2012 and responding commentaries

Designing an Effective Title

Most-read part — determines if reader even looks at paper!

- **Be specific:** state the finding, not the topic
- **Aim for 8–14 words**: enough to be precise, short enough to scan
- **Front-load key terms** for visibility
  Avoid: *"A study of. . . "*, *"The role of. . . "*, *"Effects of X on Y"*
- **Consider style:** declarative sentence, question, or noun phrase
  A noun phrase has no main verb (essentially a sophisticated label).
  "Temperature dependence of..." names phenomenon without asserting anything about it.
- **Catchy, clear, concise**[5]

Weak	*A Study of predator foraging behavior in an aquatic system*
Stronger	*Temperature mediates Type II functional responses in a dragonfly–Daphnia predator–prey system*
Strongest	*Temperature dependence of attack rates but not handling times in a dragonfly–Daphnia predator–prey system*

[5] See Singh Chawla (2025)

Structuring an Effective Abstract

1–2 sentences	Broad introduction to the field, accessible to any scientist
2–3 sentences	Detailed background for scientists in related disciplines
1 sentence	The specific problem this study addresses
1 sentence	The main result — use *"here we show"* or equivalent
2–3 sentences	What the result reveals relative to prior understanding; what it adds to existing knowledge
1–2 sentences	Broader context and implications

Key Principles[6]

- Self-contained: interpretable without access to the full paper
- Concise: limit to essential details and key message
- Include quantitative results where possible (e.g., *"attack rates declined by 43% at 10° C relative to 20° C, with handling time showing no change."*)
- State implications explicitly: do not leave the reader to infer them

[6] See *Nature-Summary-Paragraph.pdf*

Write Concisely

Brevity signals confidence in argument — length can signal the opposite

Principles

- **One idea per sentence;** one purpose per paragraph
- **Cut throat-clearing:** 1st sentence can often be deleted
- **Prefer the shorter word:** *"use"* not *"utilize"*; *"show"* not *"demonstrate"*
- **Avoid redundancy:** *"completely eliminate"*, *"end result"*, *"past history"*

A Useful Test

After drafting, ask: *does every sentence earn its place?* If a sentence neither advances the argument nor provides essential context, delete it.

Verbose	*"It is important to note that the results of the analysis that was conducted clearly demonstrate that. . . "*
Concise	*"The analysis shows that. . . "*

Discussion Writing: Implications and Speculation

The Discussion is Where Science Moves Forward

Stating only what you found is not enough. Readers need to know what your findings *mean* and where they lead.

A strong discussion . . .

- **Answers the question posed in the Introduction** — return explicitly to hypothesis or objective
- **Compares findings to prior work** — not just *"consistent with"* but *why* agreement or disagreement matters
- **States implications directly** — do not leave the reader to infer
- **Licenses informed speculation** — clearly flagged (*"we suggest. . . "*, *"one possibility is. . . "*) but not avoided
- **Identifies limitations honestly** without undermining the contribution
- **Opens new questions** — the best papers leave a field more curious than they found it

Watson & Crick (1953): *"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."* Had they omitted this implication, acceptance of the model would have been substantially delayed.

Figure-Driven Storytelling

Figures Are Not Decoration

A well-designed figure conveys structure, mechanism, or pattern that would require thousands of words to describe — and is more memorable.

- **Plan figures before writing:** if you can sketch your story as a sequence of figures, your narrative structure is sound
- **Each figure should make one clear claim** — if it requires extensive caption explanation, simplify
- **Figures and text must be mutually reinforcing,** not redundant — do not describe in prose what is already visible in the figure
- **Captions should be self-contained:** a reader should understand the figure without reading the main text
- **Order figures to mirror narrative arc** — the figure sequence is a parallel story

Use figures creatively and purposefully

Conceptual diagrams	Hypotheses, frameworks, food webs
Data figures	Results; patterns across treatments or gradients
System figures	Study site, organism, or experimental design
Synthesis figures	Meta-analytic summaries; effect size plots

Managing Jargon and Terminology

Jargon is a double-edged sword
Exists to enable precision, not to signal membership. Overuse narrows audience, obscures weak reasoning, and slows reader.

- **Define terms on first use** — even for a specialist audience
- **Use abbreviations sparingly** — only for terms used $\gtrsim 5$ times
- **Calibrate to audience** — what is jargon for general reader may be essential shorthand for specialist
- **Be especially careful in:** the Abstract (broadest readership), Introduction (context-setting), and Discussion (implications must be accessible)

Jargon-heavy	*"Spatiotemporal heterogeneity in resource availability mediates trophic cascade attenuation across productivity gradients."*
Clearer	*"Patchy resource availability weakens top-down control by predators in more productive systems."*

Analysis of Effective Papers

- Identify patterns in engaging papers
- Note what works and why
- Analyze overall structure and organization:
  - How do authors present their problem?
  - What background information do they provide?
  - How do they organize results?
  - How do they connect findings to broader context?
- Examine writing:
  - Sentence construction and clarity
  - Use of active vs. passive voice
  - Paragraph organization and transitions
  - Balance of technical detail and accessibility
- Model your own writing on effective examples

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Overview

The Literature Problem

Two Competing Demands

- Knowing the **foundational and classic** literature in your field
- Staying current with a **rapidly expanding** body of new work — often across multiple subfields

Common Failure Modes

- **Passive accumulation:** PDF hoarding without reading or annotation
- **Over-reading without synthesis:** consuming papers without integrating them into an argument
- **Reliance on incomplete system:** searching only Google Scholar, relying only on specific keyword alert

Key Principle

You do not need to read everything. You need a **structured system** for filtering, organising, and revisiting literature that balances skimming and deep reading.

Overview

Staying Current: Alerts and Feeds

Set up alerts — then let new literature come to you

Journal ToC / bioRχv Alerts Subscribe through websites or aggregate via RSS reader (e.g., Feedly). Best for core journals in your subfield — gives you everything, not just keyword matches.

Web of Science / Google Scholar Alerts Key paper and keyword alerts delivered by email on a schedule you control. *Highly filtered; low noise.*

ResearchGate / Google Scholar Profile Suggestions Follow authors in your field, surface papers based on your reading history.

Social Media Direct announcements of new papers; hashtags and community accounts surface work quickly. *Informal but fast.*

Strategy

Separate **scanning** from **deep reading**; schedule dedicated time for each.

Staying Current: Active Filtering

Three-Tier Filtering System

1. **Scan:** titles / abstracts only — decide relevance in under 60 sec.
2. **Triage:** save potentially relevant PDFs to reference manager or folder
3. **Deep Read:** full reading reserved for highest-priority papers

Heuristics for Prioritisation

- Direct relevance to current or future projects
- Methods, datasets, or frameworks of potential use
- Intriguing novelty / fun

AI-Assisted Reading Tools (examples)

NotebookLM	Query PDFs conversationally; rapid orientation to new subfield
Speechify	Convert papers to audio for assessment in low-focus time

Reading everything is inefficient

Selective re-reading is where understanding develops.

Overview

Literature Search Strategies (Traditional)

Citation Chaining

- **Backward search:** scan for common references in recent papers
- **Forward search:** scan and follow citations of key papers

Author Search

- Identify 5–10 leading (frequently cited) researchers; track the citation records of most cited papers

Start Broad, Then Narrow

- Use general keywords to identify core papers; refine iteratively using Boolean operators (`AND`, `OR`, `NOT`)
- Update as terminology, useful keywords and authors become clearer

Goal (especially for new-to-you fields)

Identify the **core intellectual structure** of a topic, not just a list of papers.

Literature Search Tools

Web of Science Curated index; strong citation tracking and bibliometric tools; advanced Boolean searches and exportable results. The Standard for systematic searches. *Requires institutional access.*

Google Scholar Far less precise than WoS but broad coverage including preprints, theses, and grey literature. Useful for quick searches and obtaining PDFs. *Free.*

AI-Assisted Tools Tools evolving rapidly. Reliability growing but TBD.

- **Elicit:** structured queries and paper summaries
- **Scite:** identifies whether papers *support* or *contrast* the work they cite
- **Semantic Scholar / ResearchRabbit:** relevance ranking, topic mapping, and citation network visualisation

Overview

Reference Managers

Core Function

Database to organize sources and auto-format citations and bibliographies

Other Functions

- Enable tagging, search, and annotation
- Store and organise PDFs
- Sync across devices; shared group libraries

Zotero[7] (most used?)

- Integrates with Word, Google, LaTeX (w/ `Better BibTeX`)
- Browser plugin captures metadata and PDFs in one click
- Group libraries: share collections with co-authors in real time
- Supports tags, notes, related items, PDF annotation

Alternatives include: BibDesk[1] (LaTeX), Mendeley[2] (strong PDF reader), EndNote[2] (institutional standard), Paperpile[2] (Google Docs integration)

[7] Free (maybe limited) and open-source; [2] **$$$** proprietary

Organizing Sources

Build Structure, Not Just Storage

- Organise by **project** or **manuscript**, not by topic alone — projects have deliverables
- Use **tags** for cross-cutting themes (e.g., `classic`, `2read`, `cited-in-ch2`)

Recommended Approach: Two-Layer System

- **Reference manager:** storage, PDF access, citation generation
- **External notes** (e.g., a plain-text or Markdown file per project): conceptual synthesis, argument mapping, connections across papers

The two layers complement each other

The reference manager stores, the notes file thinks

Annotated Bibliographies

Purpose

- Convert reading into structured, reusable knowledge
- Force active engagement (you can't annotate what you've only skimmed)
- Becomes raw material for your Introduction and Discussion sections

What to Include per Entry

- Core question or objective of the paper
- Key findings and conclusions
- Methods or analytical approach
- Relevance to your own work
- Limitations or unresolved gaps

Suggestions

- Write in your own words — not copy-pasted from abstract
- Keep entries concise: 3–6 bullet points
- Use Zotero's note field to keep with reference
- Update entries as your understanding evolves

Overview

Integrating Literature in Your Writing

Writing with specific papers in mind

- *Pros:* precise citation; accurate representation of prior work; anchors claims to evidence
- *Cons:* can produce fragmented, citation-heavy prose; limits synthesis; writing follows the literature rather than your argument
- *Best for:* methods sections, direct comparisons, specific statistics

Writing without specific papers in mind

- *Pros:* promotes conceptual clarity; argument-driven structure; encourages synthesis, faster drafting[8]
- *Cons:* risk of vague/unsupportable claims; requires well-stocked library; misrepresented citations
- *Best for:* introduction and discussion (synthesis)

[8] *"Write drunk, edit sober."* — Hemingway

Best Practice: The Hybrid Writing Strategy

Step 1 — Draft Conceptually

- Write the argument without citations
- Focus on logic, structure, and your own voice
- Placeholder (`[CITE]`) where citation is needed

Step 2 — Map Literature Onto the Argument

- Add citations to support each claim
- Replace general statements with specific evidence
- Consult annotated bibliography — most sources are likely there

Step 3 — Refine for Balance

- Avoid over-citation — not every sentence needs a reference
- Ensure key and foundational works are represented
- Verify every citation against the source
- Never cite a paper you have not fully read

Recommendation

Use argument-first writing for structure and flow, then verify every citation.

Overview

Takeaway

Knowing the classics and staying on top of the literature is a **process** and is a part of your **job** as a scientist (and educator).

Recognize it as such and **regularize it** as part of your weekly schedule.

Remember the four pillars:

Discover: alerts + citation chaining + active search

Filter: scan → triage → deep read

Organise: reference manager + external synthesis notes + annotated bibliography

Integrate: argument-first drafting → citation mapping → refinement

Takeaway

Managing the literature is not about volume.
It's about **structure, selectivity, and synthesis**.

IB 514: Scientific Writing

All course slides strung together

March 20, 2026

Dissertation Manuscripts as Collaborative Efforts

- **You own the draft.** As first author, you are responsible for producing and maintaining the manuscript — not for waiting on co-author input before moving forward.
- **Your advisor may be a co-author, not an editor.** They may revise and redirect, but the intellectual initiative should come from you.
- **Co-authorship is not automatic.** Contribution, not relationship or seniority, should determine authorship; clarify expectations *before* writing begins.
- **Co-authors have competing priorities.** You, your advisor, collaborating lab member, collaborator at another institution, need not share the same incentives.

Tools for Collaborative Writing

Choose tools that match your collaboration needs

MS Word / Apple Pages Share via email or cloud storage. Well-known; track changes; offline editing; weak for complex or equation-heavy formatting.

Google Docs Real-time cloud-based editing. Low barrier to entry; suggesting mode; comment threads; best for early drafts or non-LaTeX collaborators.

Overleaf Cloud-based LaTeX editor with real-time collaboration. Comment threads; full revision history; Git sync; the standard for LaTeX manuscripts with multiple authors.

Git / GitHub Version-controlled repositories; cloud-based. Full audit trail; branching; ideal for LaTeX and code; not real-time; steeper learning curve.

Practical Rule: Agree on a tool *before* writing begins. Mid-project migrations are time-consuming and introduce errors.

File Versioning

If you're not using cloud-based or version control software...

File Naming Ambiguous filenames are a form of technical debt. Adopt a convention and enforce it from day one.

Avoid	`draft_final_v2_JSmith_revised_FINAL.docx`
Better	`2025-11-14_manuscript_v07.tex`

Versioning Conventions

- Use **ISO dates** (`YYYY-MM-DD`) or **sequential version numbers** (`v01`, `v02`) — not both. Append commenting author initials.
- **Tag submission-ready versions** explicitly (`v_submitted`, `v_R1`)
- **Don't delete old versions** — archive them in an `/archive` subfolder

Comment and Annotation Discipline

- Resolve or explicitly defer all comments before circulating a new draft
- Use GitHub Issues or Google Docs comment threads to track unresolved questions and decisions
- Do not leave unresolved comments at submission — some journal portals expose tracked changes to reviewers

Establishing Norms and Expectations

Collaboration is Communication

Most conflicts stem from unmet unstated expectations

Clarify at the start:

- **Authorship:** Confirm whether the advisor will be last author and who is to be offered (the right to decline) authorship.
- **Level of involvement:** Will advisor/collaborators co-write sections, provide key figures, or provide feedback on drafts?
- **Escalation path:** Who has final say on journal choice, framing, interpretation, etc.? What happens if co-authors disagree?
- **Primary channel:** email, Slack, GitHub Issues — one platform for decisions, one for quick questions
- **Response expectations:** what is a reasonable turnaround for draft review (1 week? 3 weeks?) and for quick questions (48 hours?)
- **Meeting cadence:** regular check-ins vs. milestone-driven meetings — both have costs and benefits

Decision documentation: For large collaborations; all substantive decisions recorded in a shared log (email thread, shared doc, etc.)

Common Issues

Common Friction Points

Advisor rewrites your draft: Treat as mentorship; study the changes; ask for rationale on substantive edits.

Co-author delays for weeks: Set a gentle deadline when providing updated draft — *"Could you get it back to me by X."*

Co-authors disagree: Raise the conflict openly; do not triangulate between them; request a joint discussion.

A Common Failure Mode

Decisions made verbally in hallway conversations or lab meetings that are not recorded. Three months later, co-authors remember different outcomes.

Resolving Disagreements

Identify the Type of Disagreement First

Type	Example	Possible Resolution Pathways
Scientific	Disagreement over statistical model choice or interpretation of results	Resolve with evidence; consider additional analyses; consult a neutral expert
Framing	Disagreement over how to position paper's contribution	Consult target journal scope; summarize framing of comparable studies
Stylistic	Disagreement over sentence-level writing choices	Defer to the lead author; style is not worth extended negotiation
Interpersonal	Conflict rooted in prior relationship dynamics, credit concerns, or workload resentment	Address directly and early; do not let it contaminate scientific decisions

- Separate *positions* (what someone wants) from *interests* (why they want it) — interests are usually reconcilable
- Put manuscript's quality and reader's needs at the center of dispute
- Seek advice from neutral party (e.g., committee member) if direct resolution is failing

Hierarchy and Power Imbalance

Common Sources of Imbalance

- **Career stage:** harder for student to push back on a advisors framing choices, even when student is domain expert
- **Disciplinary norms:** fields may have incompatible assumptions about authorship order, target journal status, publication timeline
- **Publication pressure:** misaligned incentives around when and where to publish can fracture teams

Power Differentials Are Structural, Not Personal

Collaborations can involve genuine differences in career vulnerability and publication pressure. Ignoring these dynamics does not neutralize them.

Protective Strategies

- Document your contributions in writing as work proceeds — do not rely on memory or goodwill at submission time
- Raise authorship expectations explicitly and early, before significant work has been invested
- Consider drafting the *Authorship Contributions* statement early

Dividing and Tracking Contributions

Ambiguity about responsibility is a project risk

Task Delegation Principles

- **Assign ownership, not just involvement** — one person is responsible for each deliverable, even if others contribute
- **Be explicit about scope:** *"draft the methods section"* is a task; *"help with methods"* is not
- **Revisit assignments** as the project evolves — initial task maps rarely survive contact with data
- **Distinguish writing tasks from research tasks:** the person who ran the analysis may not be the best person to write it up, and vice versa

Contribution Tracking Tools

Google Sheets Simple shared task list with owner, status, and deadline columns — low barrier, universally accessible.

GitHub Issues Create tasks with milestone tracking; integrates directly with version-controlled repositories.

Asana/Trello/Notion Visual project boards suited to larger or longer-running collaborations.

The Authorship Conversation

When to Have It

Early — ideally before data collection begins, certainly before writing starts. Revisit at submission. (Conversations become exponentially harder once significant effort has been invested.)

What to Establish

- Who will be an author vs. acknowledged contributor?
- What is the expected authorship order and on what basis?
- What happens if contributions change substantially during project?
- Who is the senior and the corresponding author?
- What is the plan if a co-author becomes unresponsive before submission?

Avoid

- **Gift authorship:** including advisor/collaborator who did not contribute substantially/intellectually
- **Ghost authorship:** omitting those who contributied substantially
- **Retroactive negotiation:** changing authorship order after submission without full co-author agreement

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The Peer Review Pipeline

What happens after you hit submit?

1. **Submission** Manuscript enters the journal's editorial system; author receives an acknowledgement and manuscript ID.
2. **Editorial Triage** Editor-in-Chief or handling editor assesses scope fit and minimum quality threshold. Desk rejection occurs often within days without external review.
3. **Reviewer Invitation** Editor invites 2–4 reviewers; many (most) decline. This stage can take days to several weeks.
4. **External Review** Reviewers evaluate the manuscript and return written reports, within 2–8 weeks (often longer in practice).
5. **Editorial Decision** Editor synthesises reviews into one of four (or more) outcomes: **accept**, **minor revision**, **major revision**, or **reject** (with or without prejudice).
6. **Revision & Resubmission** Authors revise and submit response letter.
7. **Acceptance & Production** Manuscript enters copyediting, typesetting, and proofing before online publication.

Typical timeline: first decision in 3–5 months; full acceptance in 6–18 months across all revision rounds.

The Editor's Perspective

What editors are looking at

- **Scope:** does the manuscript fit the journal's aims and readership?
- **Significance:** does it advance the field enough to justify publication?
- **Novelty:** is it sufficiently distinct from recent literature?
- **Presentation:** is the manuscript professionally prepared?

Why desk rejection happens

- Obvious scope mismatch
- Insufficient (perceived) methodological rigour on first read
- Incomplete submissions (missing figures, references, etc.)
- Manuscript prepared for a different journal (wrong format, wrong citation style)

What this means for you as an author

- Tailor your cover letter to the journal
- Polished submissions reduces editorial hesitation
- Desk rejection is typically not a judgement of your science — it is usually a targeting problem

The Reviewer's Perspective

What reviewers evaluate

- Conceptual framing, novelty, significance of research question
- Appropriateness and rigour of methods
- Validity of results and statistical analyses
- Interpretation: do conclusions follow from the data?
- Presentation: clarity, figures, writing quality
- Completeness of citations and engagement with prior work

What reviewers shouldn't be required to do

- Editing your manuscript for you
- Reading a poorly organised paper charitably

Reviewers are busy volunteers

- Clear, well-structured manuscripts reduce reviewer burden and tend to produce more constructive feedback
- Reviewer fatigue is real: dense, difficult manuscripts are evaluated less generously

Tip: as you revise your writing, ask yourself — *am I making it easy for a tired expert to understand and trust this work?*

The Author's Perspective

Accept Rare on first submission. Minor corrections likely requested at proofing stage.

Minor Revision The manuscript is broadly acceptable; targeted changes required. Typical turnaround: 2–4 weeks.

Major Revision Substantial work required (new analyses, restructuring, additional experiments). Usually one more full review round. Typical turnaround: 4–12 weeks. *This is the most common outcome.*

Reject The manuscript will not be reconsidered at this journal.[9] May include encouragement to resubmit elsewhere.

[9] Entirely new manuscript expected for "rejection without prejudice."

First steps

First steps after receiving reviews

1. Read once for overall tone — then **step away**
2. Return the next day and read for content, not emotion
3. Identify which comments require new analysis, which require writing, and which require clarification only
4. Distribute your synthesis and discuss a plan with co-authors *before* revising

Rejection Management

Rejection is normal — not exceptional

- Rates can exceed 70–90%; desk rejection of 50%+ at high-impact journals
- Most published papers were rejected at least once
- Rejection does not mean the work is unpublishable — it usually means the targeting was wrong

What to do immediately after rejection

- Read the decision letter and reviews carefully — even a rejection often contains actionable feedback
- Discuss with co-authors: is the feedback *scientific* (revise before resubmitting) or *editorial* (wrong journal)?
- Identify the next target journal *before* making any revisions

Appealing: appropriate if review contains clear and foundational factual error or demonstrable misreading of the work. Rarely successful, can take weeks or months, and delays resubmission elsewhere

Rule of thumb: Rejection sucks. Get over it and aim to submit a rejected manuscript elsewhere within 4–6 weeks. **Momentum matters**.

Revision & Resubmission Workflow

Organise before you write

- Extract all editor and reviewer comments into a single numbered list
- Identify overlap and dependencies:
  - Overlap across reviewers' comments
  - Revisions requiring new analyses prior to writing
- Set a revision timeline (considering journal's request); request extension early if needed

Revise manuscript and draft response letter in parallel

- Draft response as each comment is addressed, not retrospectively
- Track changes in manuscript for co-authors
- Keep change-tracked version for editor

Verify

- Every response references a line number in the revision
- Every comment has a response (even just "*Implemented on line x.*")
- Changes haven't broken figure or table references

Responding to Reviewer Feedback

The response document is as important as the revision

- Editors often read your response *before* re-reading the manuscript — make it clear and complete
- Address **every** comment, even if you disagree
- Use a structured format: quote the comment, state your response, indicate the change made and its location[10]

Handle disagreement professionally

- You may respectfully decline a reviewer request — but you must explain why clearly and with evidence
- Phrase pushback as: *"We respectfully disagree because. . . and instead propose. . . "*
- Never argue with tone; argue with data and logic
- If two reviewers contradict each other, flag it to the editor, use it to your advantage (if appropriate), and propose a resolution

[10] See provided LaTeX template.

The Proofing Process

Proofing is a **correction stage**, not a revision stage.
Increasing done through journal's online proofing system.

Read very carefully, especially...

- All author names, affiliations, and corresponding author details
- All equation, special character, and Greek symbol rendering
- Figures and tables in the correct order, at sufficient resolution, correctly captioned

Practical notes

- Proofing deadlines are typically 48–72 hours
- Read proof against manuscript line by line, not from memory
- Uncaught errors become permanent in published record; corrections require formal erratum

Remember to celebrate!

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Pre-Submission Review by Friendly Reviewers

Selecting Reviewers

- Avoid/Seek reviewers with strong prior positions on your findings
- Prioritize people who will give *honest* feedback, not reassurance
- Consider who might make a better journal reviewer

Briefing Reviewers

- State the target journal and audience
- Set a realistic deadline (2-3 weeks is standard)
- Clarify what kind of feedback is useful: structure, argument, clarity, or all three
- Identify specific concerns you want addressed

Thank them by name in the Acknowledgments.
Consider mentioning in Cover Letter.
Friendly review is *not* a substitute for rigorous self-revision. Send out only when you believe the manuscript is essentially complete.

Final Formatting

Things I ignore:

- Font type and size
- Margin requirements
- Journal-requested citation format

Things I don't ignore:

- Word/page limits (*incl. title, abstract, main text*)
- Line spacing (*double*) & page numbering
- Figure placement (*inline except if explicitly specified*) and formats (*pdf, tiff, eps*)
- Cover page and/or end-of-manuscript requirements
  - Author contribution and Acknowledgments
  - Conflict of Interest disclosure requirements
  - Data & Code Availability statement requirements

Check:

- Triple-read for spelling and punctuation
- Citation completeness and consistency
- Journal's double-blind requirements

Author Contributions & the CRediT Taxonomy

Purpose

- Distinguish **authorship** from **acknowledgment**
- Increase transparency and accountability
- Expose *ghost authorship* and *gift authorship*
- Increasingly indexed by databases (e.g., OpenAlex)

The CRediT Taxonomy (14 Roles) `credit.niso.org`

Conceptualization	Methodology
Data Curation	Project Administration
Formal Analysis	Resources
Funding Acquisition	Software
Investigation	Supervision
Validation	Visualization
Writing — Original Draft	Writing — Review & Editing

Author Contributions Statement

Common Formats

1. **Prose (narrative)**
   "J.S. conceived the study and led data collection. M.R. performed statistical analysis. . ."

2. **Role-tagged list**
   Conceptualization: J.S., M.R.
   Formal Analysis: M.R.
   Writing–Original Draft: J.S.
   Writing–Review & Editing: All authors

Imperative: Agree on contributions *before drafting*, not retroactively. Disagreements are common, uncomfortable, and potentially costly.[11]

[11] See *Collaboration* class.

Open Science

- Promotes **transparency** and **reproducibility**
- Increases citation impact and research visibility
- Broadens engagement across the research community
- Enhances credibility

Key Considerations

- Article Processing Charges (APCs) can be substantial
- Data sensitivity and participant privacy constraints
- Institutional and funder mandates increasingly require OA

Open Access Paper Publication Models

Type	Who Pays	Description
Gold	Author / funder	Published OA immediately by the journal; APC required
Green	Nobody	Author self-archives a preprint or accepted manuscript in a repository (e.g., biorχiv)
Diamond	Nobody	Community- or institution-funded; no fees for authors or readers
Hybrid	Author	Subscription journal offers OA for individual articles upon APC payment
Bronze	Nobody	Freely readable on the publisher site *without* open licence; access can be revoked

Note: Green OA costs nothing and is compatible with most journals. Check policies at `https://openpolicyfinder.jisc.ac.uk`.

Data & Code Sharing

- Increasingly required by journals and funders
- Can be cited - Digitial Object Identifier (DOI)
- Common repositories:
  - Data: Zenodo, FigShare, Data One
  - Code: GitHub, Zenodo
  - Supplementary materials (*no longer considered enough*)
- Best practices:
  - Include comprehensive documentation and README files
  - Provide appropriate licensing

FAIR Data Principles

FAIR defines standards to maximise reuse and reproducibility.

F — Findable Data are assigned persistent identifiers (e.g., DOIs) and described with rich, searchable metadata.

A — Accessible Data and metadata are retrievable via open, standardised protocols; access conditions are clearly stated.

I — Interoperable Data use common formats, vocabularies, and ontologies to enable integration with other datasets and tools.

R — Reusable Data carry explicit licences (e.g., CC BY) and sufficient provenance information for others to build on them.

Full guidelines at `go-fair.org/fair-principles`.

Note: FAIR does not require data to be *open* — sensitive data can be FAIR with controlled access.

Preprint Repositories: Examples

bioRχiv Life sciences; operated by Cold Spring Harbor Laboratory; the dominant biology preprint server since 2013.

EcoEvoRχiv Ecology and evolutionary biology; community-governed; supports registered reports and post-publication versions.

arχiv Physics, mathematics, computer science, quantitative biology; the oldest major preprint server (est. 1991); moderated by field endorsers.

Zenodo General-purpose; accepts preprints, datasets, code, and presentations; assigns DOIs; widely used for open science compliance.

Authorea Collaborative writing and preprint platform; supports live figures, data embedding, and inline commenting; blurs the boundary between manuscript and publication.

Preprint Repositories: Primary Pros & Cons

Advantages

- **Speed:** findings are public immediately, months before formal publication
- **Priority:** timestamped record of contribution
- **Visibility:** indexed by Google Scholar, etc.
- **Feedback:** community comments before peer review
- **Open access:** freely available
- **Funder compliance:** satisfies some open science mandates immediately

Disadvantages

- **Journal policies:** some journals discourage or prohibit
- **No peer review:** errors and weak claims are public
- **Perception:** some fields and hiring committees do not credit preprints
- **Retraction gap:** flawed preprints may persist even after journal rejection
- **Media risk:** premature press coverage before validation

Peer Community In... (PCI)

`PeerCommunityIn.org`

Goals

- Transparent peer review *before* (or *instead of*) journal submission
- Eliminate fees for authors and readers
- Reduce dependence on commercial publishers

How It Works

- Authors submit preprint (e.g., to bioRχiv), then request PCI review
- Volunteer recommender (acting editor) solicits and coordinates reviews
- Accepted preprints receive a **PCI Recommendation** — citable endorsement
- PCI-friendly journals accept recommended preprints, or regular review

figs/PCI_list.png

Page Charges & Publication Fees

- **Article Processing Charges (APCs):** \$500–\$12,000+ depending on journal/OA model
- **Page charges:** \$0–\$200 per printed page for subscription journals
- **Color figure fees:** \$300–\$1,000+ per figure; often free online-only

Who Pays — and How to Find Out

- **Ask your advisor first:** most PIs budget for publication costs through grants; this is a normal conversation to have
- **Institutional waivers:** many universities (but not OSU!) have OA publishing funds; check with library
- **Funder mandates:** NIH, Wellcome Trust, and others *require* OA and may cover fees directly
- **Request a waiver:** most journals offer hardship waivers for authors without funding — ask before assuming you must pay

Bottom line: You shouldn't be expected to pay out of pocket. Clarify funding *before* submission, not after acceptance.

Suggesting Potential Reviewers

How to Identify

- Researchers who publish directly on topic or methods
- Authors cited in manuscript (but not exclusively)
- Individuals with complementary expertise (theory, methods, system)

Who to Avoid

- Current or recent collaborators (typically last 3–5 years)
- Members of your institution or close professional network
- Advisors, committee members, or mentors

Strategic Considerations

- Aim for 4–8 suggested reviewers
- Ensure diversity (perspectives, institutions, career stage, gender)
- Avoid suggesting only highly favorable reviewers

Opposed Reviewers

- Allowed by some journals but use very sparingly and justify![12]

[12] Consider use as *"friendly" reviewer*

Submission Logistics

Explore your target journal's submission system!